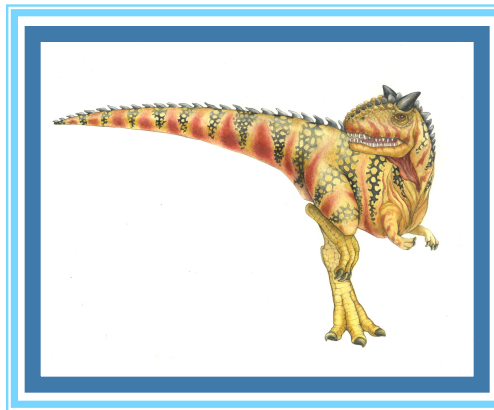


Chapter 3: Processes





Chapter 3: Processes

- Process Concept
- Process Scheduling
- Operations on Processes
- Interprocess Communication
- Examples of IPC Systems
- Communication in Client-Server Systems





Objectives

- To introduce the notion of a process -- a program in execution, which forms the basis of all computation
- To describe the various features of processes, including scheduling, creation and termination, and communication
- To explore interprocess communication using shared memory and message passing
- To describe communication in client-server systems





Process Concepts





Process Concept

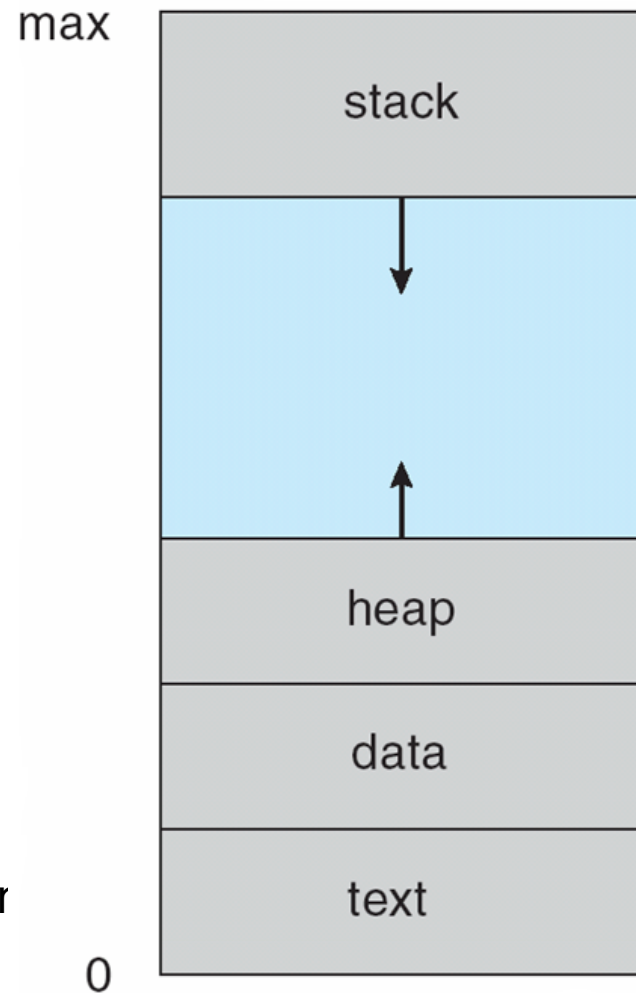
- An operating system executes a variety of programs:
 - Batch system – **jobs**
 - Time-shared systems – **user programs** or **tasks**
- **Process** – a program in execution; process execution must progress in sequential fashion
- Program is ***passive*** entity stored on disk (**executable file**), process is ***active***
 - Program becomes process when executable file loaded into memory
- One program can be several processes
 - Consider multiple users executing the same program





Process Concept (Cont.)

- Multiple parts
 - The program code, also called **text section**
 - Current activity including **program counter**, processor registers
 - **Stack** containing temporary data
 - ▶ Function parameters, return addresses, local variables
 - **Data section** containing global variables
 - **Heap** containing memory dynamically allocated during run time





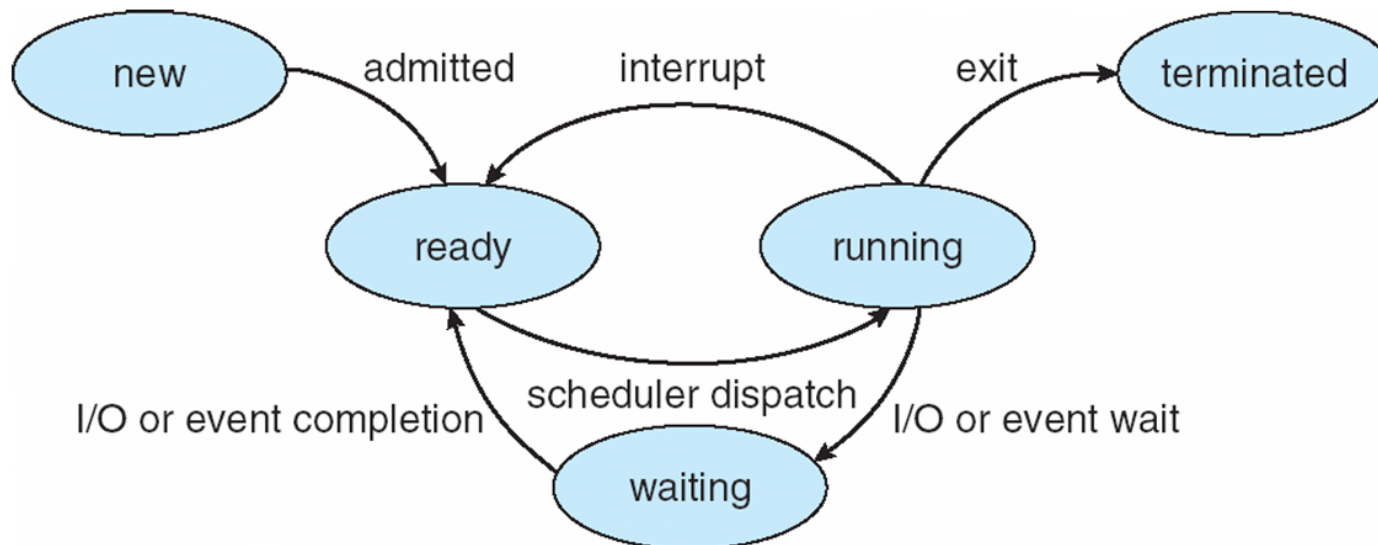
Process States





Process State

- As a process executes, it changes **state**
 - **new**: The process is being created
 - **running**: Instructions are being executed
 - **waiting**: The process is waiting for some event to occur
 - **ready**: The process is waiting to be assigned to a processor
 - **terminated**: The process has finished execution





Process Control Block

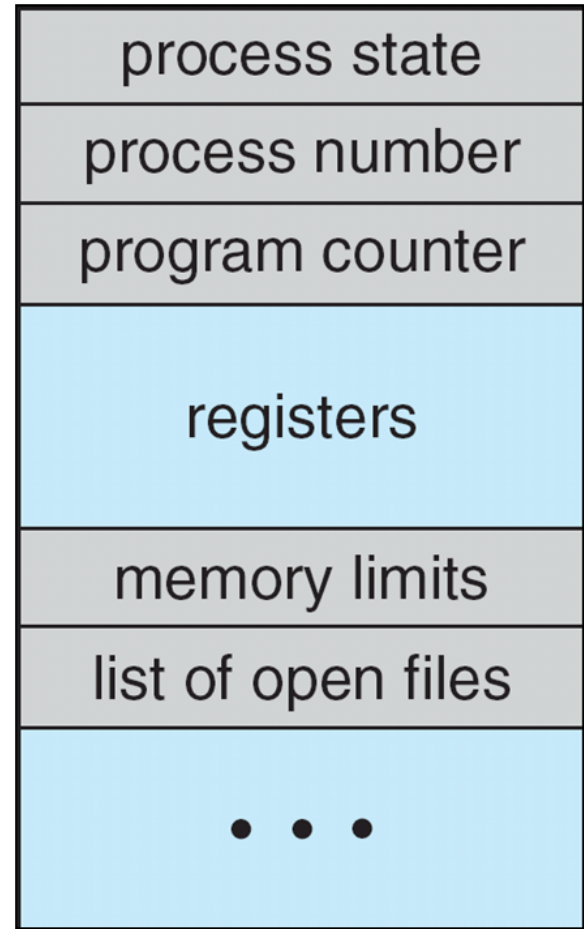




Process Control Block (PCB)

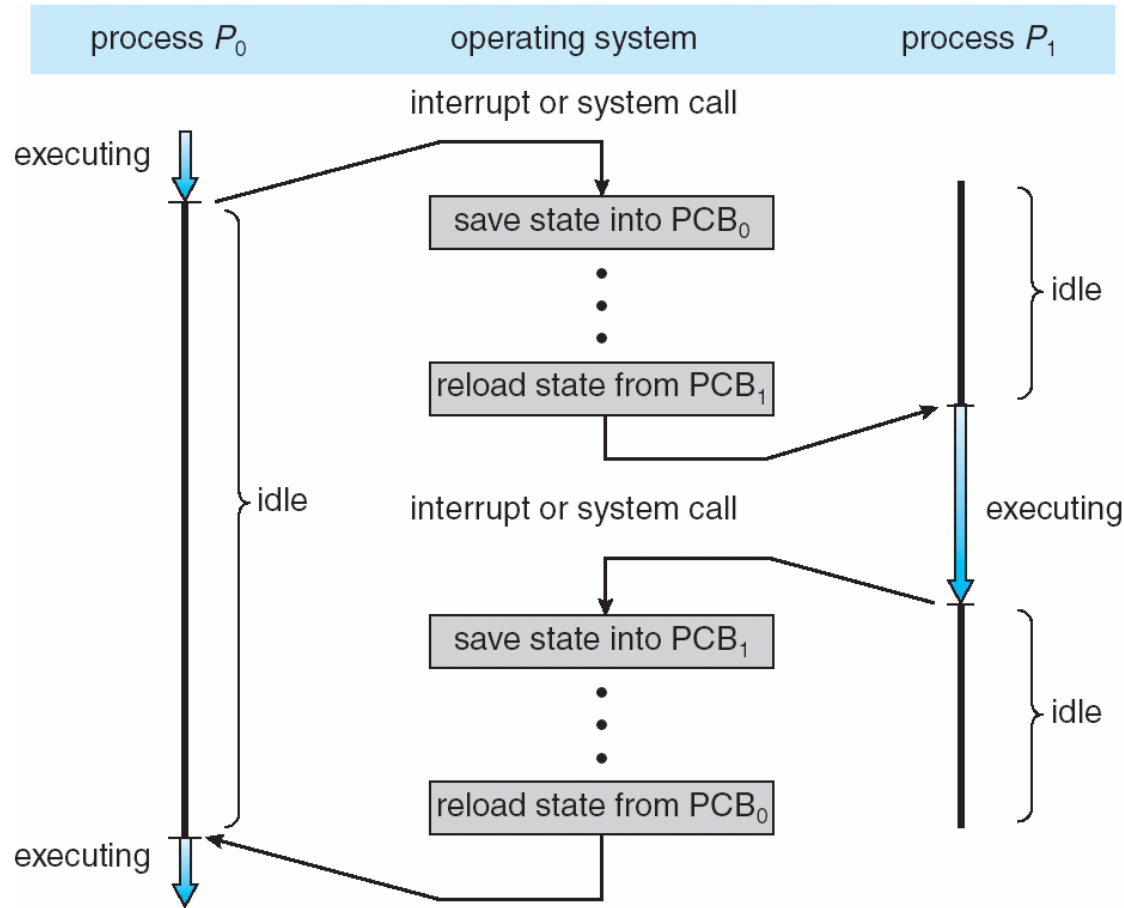
Information associated with each process
(also called **task control block**)

- Process state – running, waiting, etc
- Program counter – location of instruction to next execute
- CPU registers – contents of all process-centric registers
- CPU scheduling information- priorities, scheduling queue pointers
- Memory-management information – memory allocated to the process
- Accounting information – CPU used, clock time elapsed since start, time limits
- I/O status information – I/O devices allocated to process, list of open files





CPU Switch From Process to Process





Threads





Threads

- So far, process has a single thread of execution
- Consider having multiple program counters per process
 - Multiple locations can execute at once
 - ▶ Multiple threads of control -> **threads**
- Must then have storage for thread details, multiple program counters in PCB
- See next chapter





Process Scheduling





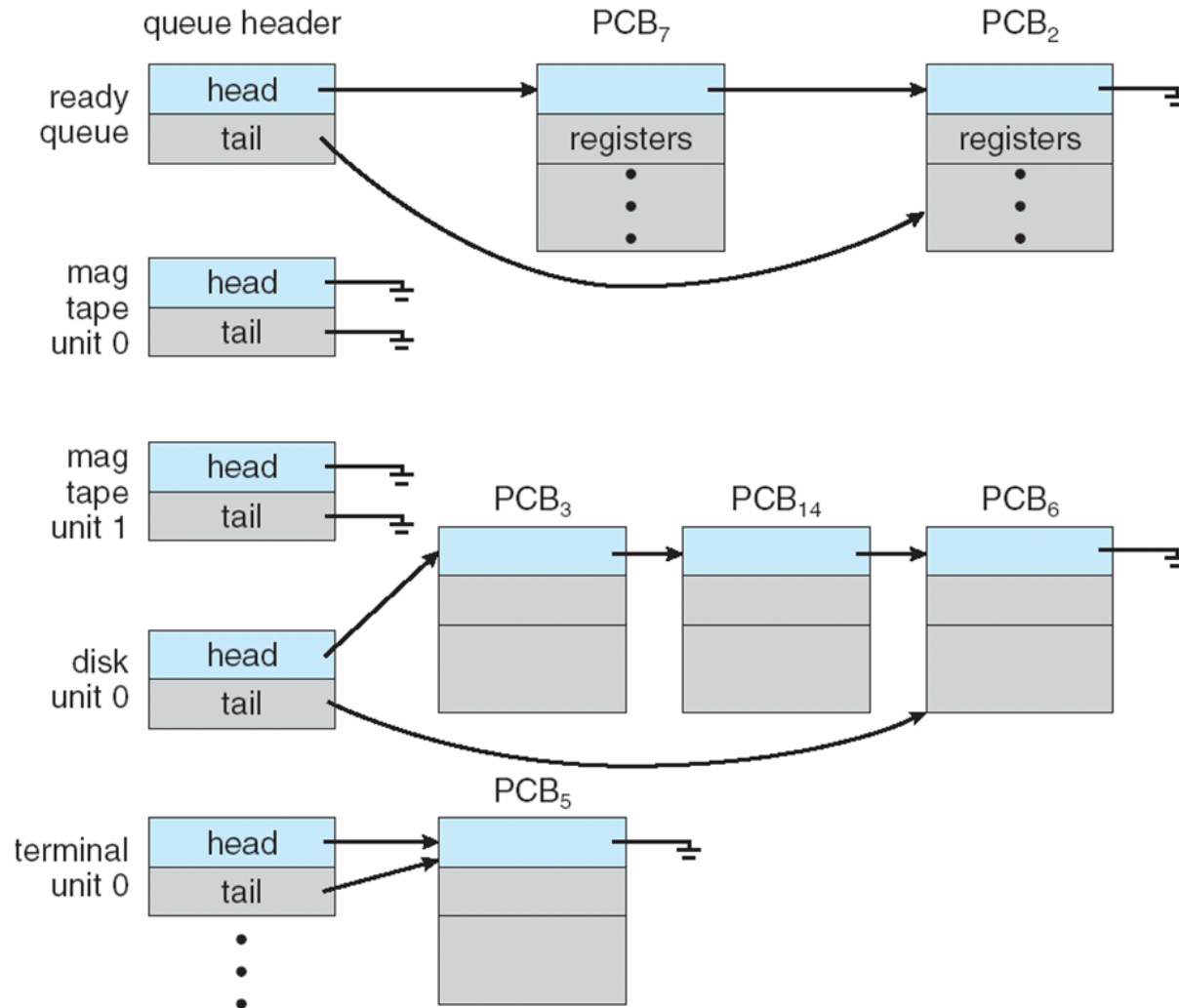
Process Scheduling

- Maximize CPU use, quickly switch processes onto CPU for time sharing
- **Process scheduler** selects among available processes for next execution on CPU
- Maintains **scheduling queues** of processes
 - **Job queue** – set of all processes in the system
 - **Ready queue** – set of all processes residing in main memory, ready and waiting to execute
 - **Device queues** – set of processes waiting for an I/O device
 - Processes migrate among the various queues





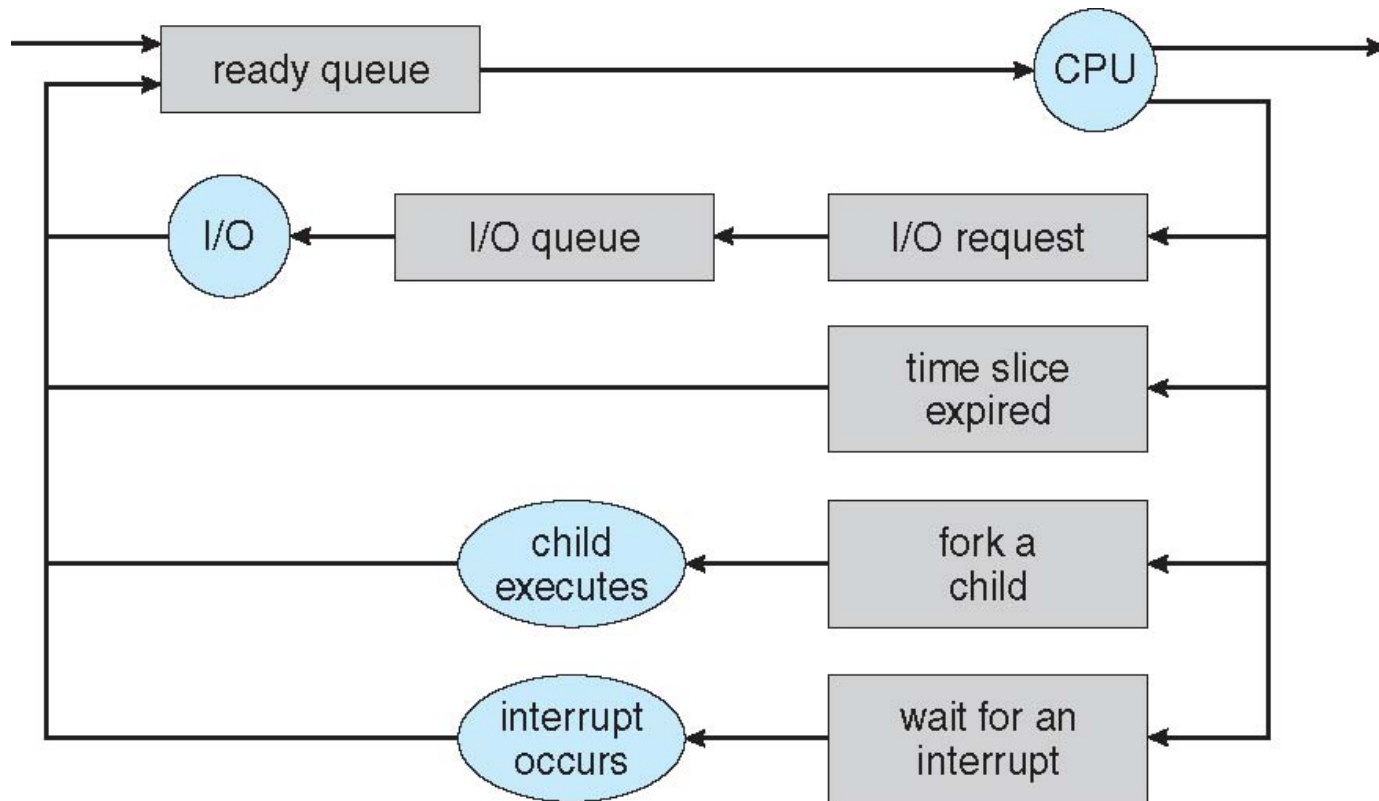
Ready Queue And Various I/O Device Queues





Representation of Process Scheduling

- **Queueing diagram** represents queues, resources, flows





Schedulers

- **Short-term scheduler** (or **CPU scheduler**) – selects which process should be executed next and allocates CPU
 - Sometimes the only scheduler in a system
 - Short-term scheduler is invoked frequently (milliseconds) $\Rightarrow$ (must be fast)
- **Long-term scheduler** (or **job scheduler**) – selects which processes should be brought into the ready queue
 - Long-term scheduler is invoked infrequently (seconds, minutes) $\Rightarrow$ (may be slow)
 - The long-term scheduler controls the **degree of multiprogramming**
- Processes can be described as either:
 - **I/O-bound process** – spends more time doing I/O than computations, many short CPU bursts
 - **CPU-bound process** – spends more time doing computations; few very long CPU bursts
- Long-term scheduler strives for good ***process mix***





Context Switching





Multitasking in Mobile Systems

- Some mobile systems (e.g., early version of iOS) allow only one process to run, others suspended
- Due to screen real estate, user interface limits iOS provides for a
 - Single **foreground** process- controlled via user interface
 - Multiple **background** processes– in memory, running, but not on the display, and with limits
 - Limits include single, short task, receiving notification of events, specific long-running tasks like audio playback
- Android runs foreground and background, with fewer limits
 - Background process uses a **service** to perform tasks
 - Service can keep running even if background process is suspended
 - Service has no user interface, small memory use





Context Switch

- When CPU switches to another process, the system must **save the state** of the old process and load the **saved state** for the new process via a **context switch**
- **Context** of a process represented in the PCB
- Context-switch time is overhead; the system does no useful work while switching
 - The more complex the OS and the PCB → the longer the context switch
- Time dependent on hardware support
 - Some hardware provides multiple sets of registers per CPU → multiple contexts loaded at once





Operations on Processes

- System must provide mechanisms for:
 - process creation,
 - process termination,
 - and so on as detailed next





Process Creation





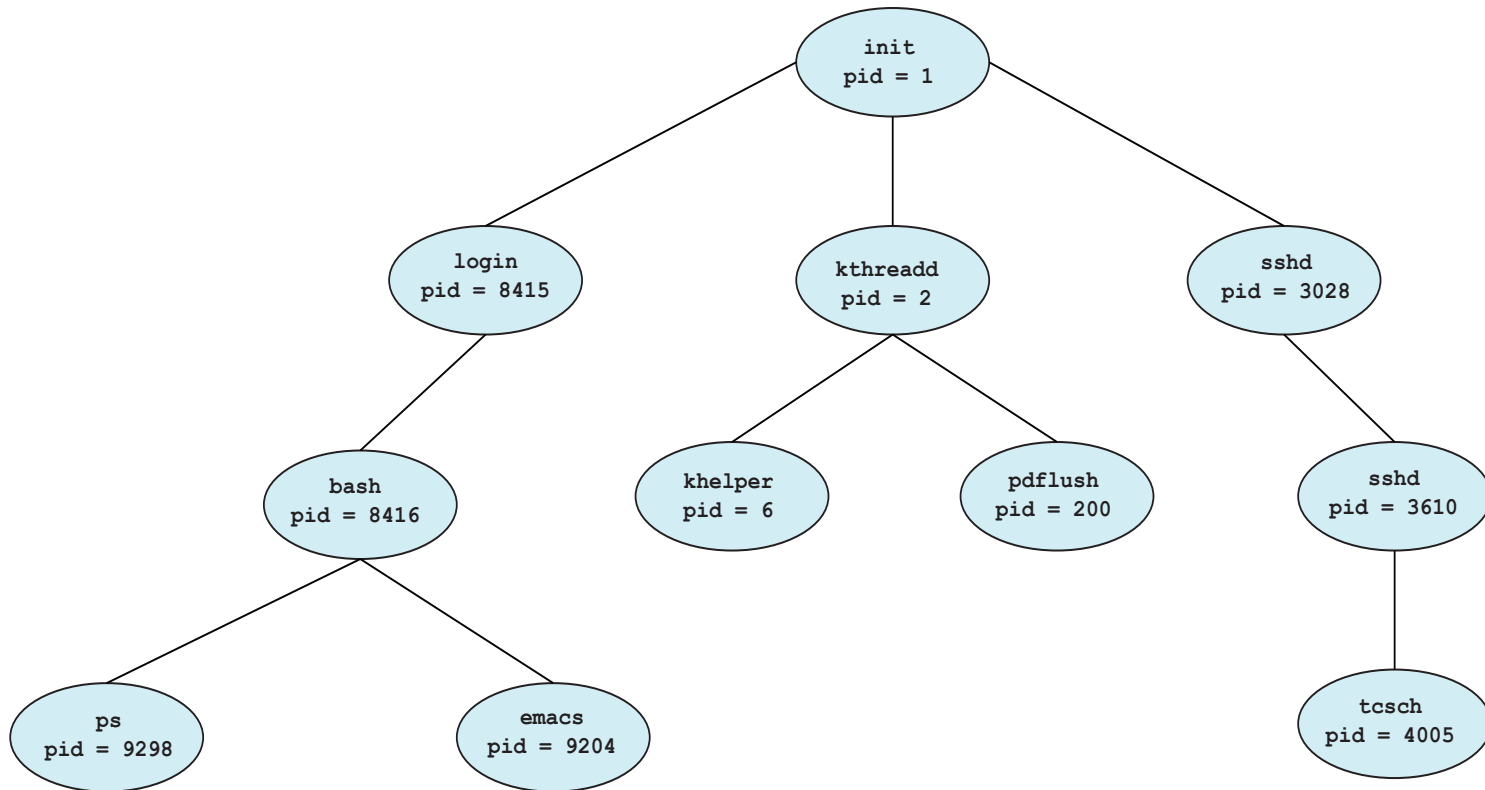
Process Creation

- **Parent** process create **children** processes, which, in turn create other processes, forming a **tree** of processes
- Generally, process identified and managed via a **process identifier (pid)**
- Resource sharing options
 - Parent and children share all resources
 - Children share subset of parent's resources
 - Parent and child share no resources
- Execution options
 - Parent and children execute concurrently
 - Parent waits until children terminate





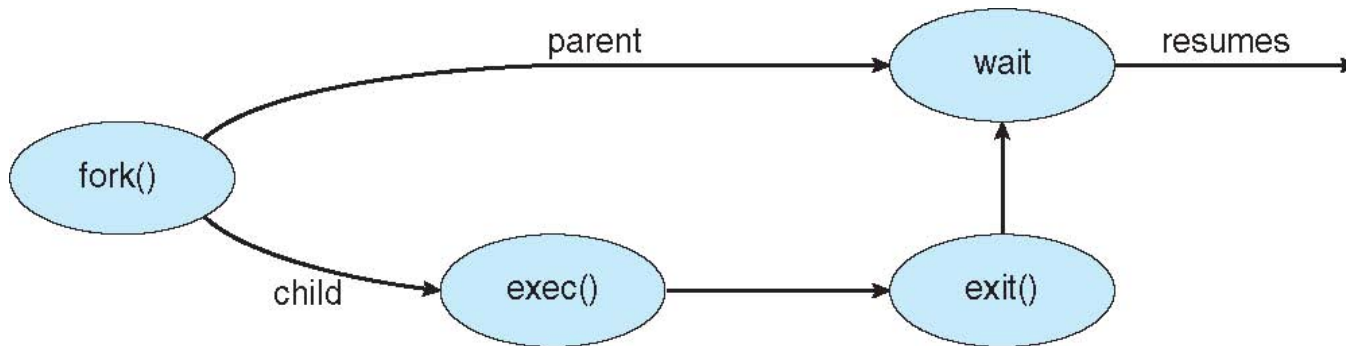
A Tree of Processes in Linux





Process Creation (Cont.)

- Address space
 - Child duplicate of parent
 - Child has a program loaded into it
- UNIX examples
 - **fork()** system call creates new process
 - **exec()** system call used after a **fork()** to replace the process' memory space with a new program





C Program Forking Separate Process

```
#include <sys/types.h>
#include <stdio.h>
#include <unistd.h>

int main()
{
    pid_t pid;

    /* fork a child process */
    pid = fork();

    if (pid < 0) { /* error occurred */
        fprintf(stderr, "Fork Failed");
        return 1;
    }
    else if (pid == 0) { /* child process */
        execlp("/bin/ls", "ls", NULL);
    }
    else { /* parent process */
        /* parent will wait for the child to complete */
        wait(NULL);
        printf("Child Complete");
    }

    return 0;
}
```





Creating a Separate Process via Windows API

```
#include <stdio.h>
#include <windows.h>

int main(VOID)
{
    STARTUPINFO si;
    PROCESS_INFORMATION pi;

    /* allocate memory */
    ZeroMemory(&si, sizeof(si));
    si.cb = sizeof(si);
    ZeroMemory(&pi, sizeof(pi));

    /* create child process */
    if (!CreateProcess(NULL, /* use command line */
        "C:\\WINDOWS\\system32\\mspaint.exe", /* command */
        NULL, /* don't inherit process handle */
        NULL, /* don't inherit thread handle */
        FALSE, /* disable handle inheritance */
        0, /* no creation flags */
        NULL, /* use parent's environment block */
        NULL, /* use parent's existing directory */
        &si,
        &pi))
    {
        fprintf(stderr, "Create Process Failed");
        return -1;
    }
    /* parent will wait for the child to complete */
    WaitForSingleObject(pi.hProcess, INFINITE);
    printf("Child Complete");

    /* close handles */
    CloseHandle(pi.hProcess);
    CloseHandle(pi.hThread);
}
```





Process Termination





Process Termination

- Process executes last statement and then asks the operating system to delete it using the `exit()` system call.
 - Returns status data from child to parent (via `wait()`)
 - Process' resources are deallocated by operating system
- Parent may terminate the execution of children processes using the `abort()` system call. Some reasons for doing so:
 - Child has exceeded allocated resources
 - Task assigned to child is no longer required
 - The parent is exiting and the operating systems does not allow a child to continue if its parent terminates





Process Termination

- Some operating systems do not allow child to exist if its parent has terminated. If a process terminates, then all its children must also be terminated.
 - **cascading termination.** All children, grandchildren, etc. are terminated.
 - The termination is initiated by the operating system.
 - The parent process may wait for termination of a child process by using the `wait()` system call. The call returns status information and the pid of the terminated process
- ```
pid = wait(&status);
```
- If no parent waiting (did not invoke `wait()`) process is a **zombie**
  - If parent terminated without invoking `wait`, process is an **orphan**





# Multiprocess Architecture – Chrome Browser

- Many web browsers ran as single process (some still do)
  - If one web site causes trouble, entire browser can hang or crash
- Google Chrome Browser is multiprocess with 3 different types of processes:
  - **Browser** process manages user interface, disk and network I/O
  - **Renderer** process renders web pages, deals with HTML, Javascript. A new renderer created for each website opened
    - ▶ Runs in **sandbox** restricting disk and network I/O, minimizing effect of security exploits
  - **Plug-in** process for each type of plug-in





# Interprocess Communication





# Interprocess Communication

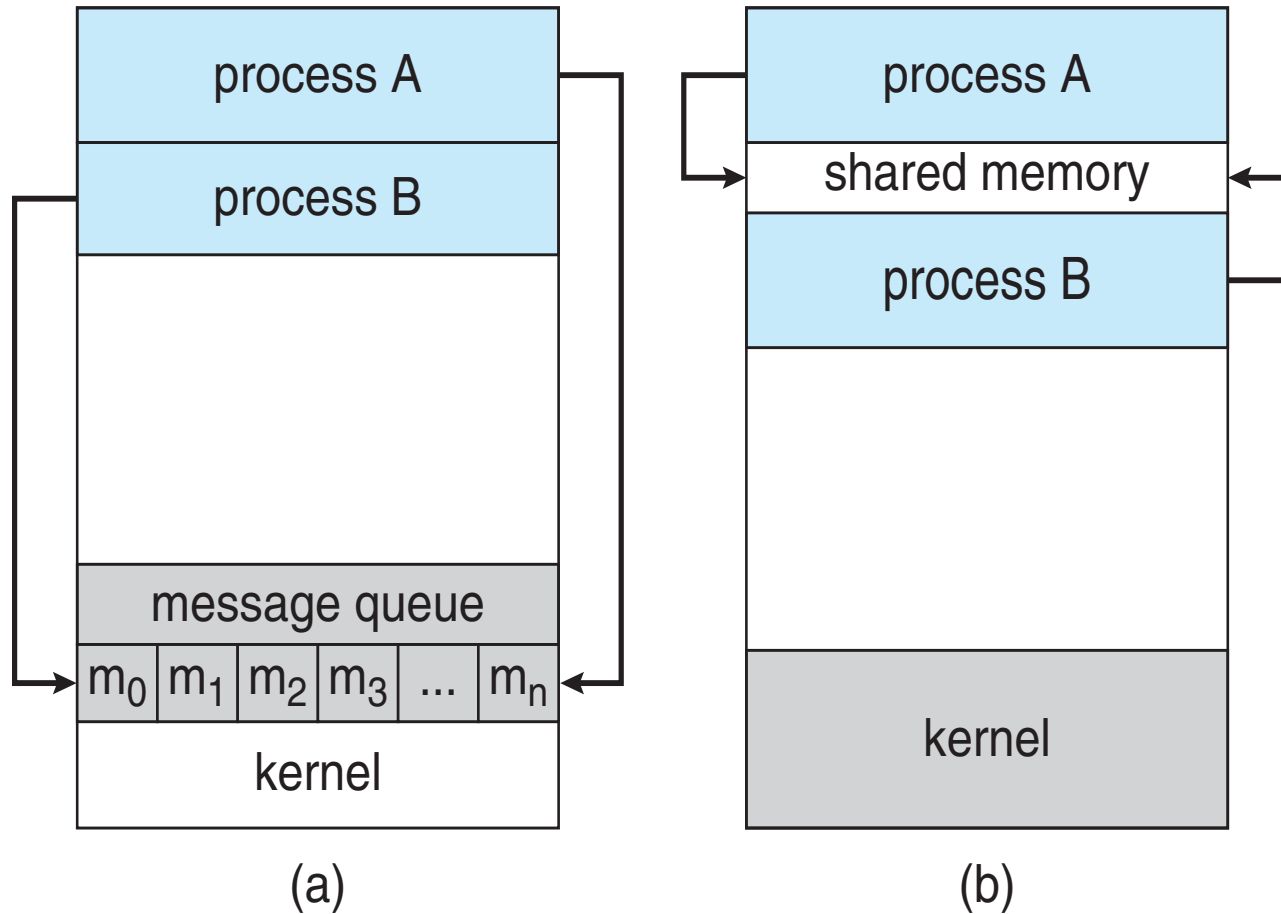
- Processes within a system may be *independent* or *cooperating*
- Cooperating process can affect or be affected by other processes, including sharing data
- Reasons for cooperating processes:
  - Information sharing
  - Computation speedup
  - Modularity
  - Convenience
- Cooperating processes need **interprocess communication (IPC)**
- Two models of IPC
  - **Shared memory**
  - **Message passing**





# Communications Models

(a) Message passing. (b) shared memory.





# Cooperating Processes

---

- **Independent** process cannot affect or be affected by the execution of another process
- **Cooperating** process can affect or be affected by the execution of another process
- Advantages of process cooperation
  - Information sharing
  - Computation speed-up
  - Modularity
  - Convenience





# Producer-Consumer Problem

---

- Paradigm for cooperating processes, *producer* process produces information that is consumed by a *consumer* process
  - **unbounded-buffer** places no practical limit on the size of the buffer
  - **bounded-buffer** assumes that there is a fixed buffer size





# Bounded-Buffer – Shared-Memory Solution

## ■ Shared data

```
#define BUFFER_SIZE 10
typedef struct {
 . . .
} item;

item buffer[BUFFER_SIZE];
int in = 0;
int out = 0;
```

## ■ Solution is correct, but can only use BUFFER\_SIZE-1 elements





# Bounded-Buffer – Producer

---

```
item next_produced;
while (true) {
 /* produce an item in next produced */
 while (((in + 1) % BUFFER_SIZE) == out)
 ; /* do nothing */
 buffer[in] = next_produced;
 in = (in + 1) % BUFFER_SIZE;
}
```





# Bounded Buffer – Consumer

---

```
item next_consumed;
while (true) {
 while (in == out)
 ; /* do nothing */
 next_consumed = buffer[out];
 out = (out + 1) % BUFFER_SIZE;

 /* consume the item in next consumed */
}
```





# Interprocess Communication – Shared Memory

---

- An area of memory shared among the processes that wish to communicate
- The communication is under the control of the users processes not the operating system.
- Major issues is to provide mechanism that will allow the user processes to synchronize their actions when they access shared memory.
- Synchronization is discussed in great details in Chapter 5.





# Interprocess Communication – Message Passing

---

- Mechanism for processes to communicate and to synchronize their actions
- Message system – processes communicate with each other without resorting to shared variables
- IPC facility provides two operations:
  - **send**(*message*)
  - **receive**(*message*)
- The *message* size is either fixed or variable





## Message Passing (Cont.)

- If processes  $P$  and  $Q$  wish to communicate, they need to:
  - Establish a **communication link** between them
  - Exchange messages via send/receive
- Implementation issues:
  - How are links established?
  - Can a link be associated with more than two processes?
  - How many links can there be between every pair of communicating processes?
  - What is the capacity of a link?
  - Is the size of a message that the link can accommodate fixed or variable?
  - Is a link unidirectional or bi-directional?





# Message Passing (Cont.)

---

- Implementation of communication link
  - Physical:
    - ▶ Shared memory
    - ▶ Hardware bus
    - ▶ Network
  - Logical:
    - ▶ Direct or indirect
    - ▶ Synchronous or asynchronous
    - ▶ Automatic or explicit buffering





# Direct Communication

---

- Processes must name each other explicitly:
  - **send** ( $P$ , *message*) – send a message to process  $P$
  - **receive**( $Q$ , *message*) – receive a message from process  $Q$
- Properties of communication link
  - Links are established automatically
  - A link is associated with exactly one pair of communicating processes
  - Between each pair there exists exactly one link
  - The link may be unidirectional, but is usually bi-directional





# Indirect Communication

---

- Messages are directed and received from mailboxes (also referred to as ports)
  - Each mailbox has a unique id
  - Processes can communicate only if they share a mailbox
- Properties of communication link
  - Link established only if processes share a common mailbox
  - A link may be associated with many processes
  - Each pair of processes may share several communication links
  - Link may be unidirectional or bi-directional





# Indirect Communication

---

## ■ Operations

- create a new mailbox (port)
- send and receive messages through mailbox
- destroy a mailbox

## ■ Primitives are defined as:

**send**(*A*, *message*) – send a message to mailbox *A*

**receive**(*A*, *message*) – receive a message from mailbox *A*





# Indirect Communication

## ■ Mailbox sharing

- $P_1$ ,  $P_2$ , and  $P_3$  share mailbox A
- $P_1$  sends;  $P_2$  and  $P_3$  receive
- Who gets the message?

## ■ Solutions

- Allow a link to be associated with at most two processes
- Allow only one process at a time to execute a receive operation
- Allow the system to select arbitrarily the receiver. Sender is notified who the receiver was.





# Synchronization

- Message passing may be either blocking or non-blocking
- **Blocking** is considered **synchronous**
  - **Blocking send** -- the sender is blocked until the message is received
  - **Blocking receive** -- the receiver is blocked until a message is available
- **Non-blocking** is considered **asynchronous**
  - **Non-blocking send** -- the sender sends the message and continue
  - **Non-blocking receive** -- the receiver receives:
    - A valid message, or
    - Null message
- Different combinations possible
  - If both send and receive are blocking, we have a **rendezvous**





# Synchronization (Cont.)

---

## ■ Producer-consumer becomes trivial

```
message next_produced;
while (true) {
 /* produce an item in next produced */
 send(next_produced);
}
```

```
message next_consumed;
while (true) {
 receive(next_consumed);

 /* consume the item in next consumed */
}
```





# Buffering

- Queue of messages attached to the link.
- implemented in one of three ways
  1. Zero capacity – no messages are queued on a link.  
Sender must wait for receiver (rendezvous)
  2. Bounded capacity – finite length of  $n$  messages  
Sender must wait if link full
  3. Unbounded capacity – infinite length  
Sender never waits





# Examples of IPC Systems - POSIX

---

## ■ POSIX Shared Memory

- Process first creates shared memory segment

```
shm_fd = shm_open(name, O_CREAT | O_RDWR, 0666);
```

- Also used to open an existing segment to share it
- Set the size of the object

```
ftruncate(shm_fd, 4096);
```

- Now the process could write to the shared memory

```
sprintf(shared_memory, "Writing to shared
memory");
```





# IPC POSIX Producer

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <fcntl.h>
#include <sys/shm.h>
#include <sys/stat.h>

int main()
{
 /* the size (in bytes) of shared memory object */
 const int SIZE = 4096;
 /* name of the shared memory object */
 const char *name = "OS";
 /* strings written to shared memory */
 const char *message_0 = "Hello";
 const char *message_1 = "World!";

 /* shared memory file descriptor */
 int shm_fd;
 /* pointer to shared memory object */
 void *ptr;

 /* create the shared memory object */
 shm_fd = shm_open(name, O_CREAT | O_RDWR, 0666);

 /* configure the size of the shared memory object */
 ftruncate(shm_fd, SIZE);

 /* memory map the shared memory object */
 ptr = mmap(0, SIZE, PROT_WRITE, MAP_SHARED, shm_fd, 0);

 /* write to the shared memory object */
 sprintf(ptr,"%s",message_0);
 ptr += strlen(message_0);
 sprintf(ptr,"%s",message_1);
 ptr += strlen(message_1);

 return 0;
}
```





# IPC POSIX Consumer

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <sys/shm.h>
#include <sys/stat.h>

int main()
{
 /* the size (in bytes) of shared memory object */
 const int SIZE = 4096;
 /* name of the shared memory object */
 const char *name = "OS";
 /* shared memory file descriptor */
 int shm_fd;
 /* pointer to shared memory object */
 void *ptr;

 /* open the shared memory object */
 shm_fd = shm_open(name, O_RDONLY, 0666);

 /* memory map the shared memory object */
 ptr = mmap(0, SIZE, PROT_READ, MAP_SHARED, shm_fd, 0);

 /* read from the shared memory object */
 printf("%s", (char *)ptr);

 /* remove the shared memory object */
 shm_unlink(name);

 return 0;
}
```





# Examples of IPC Systems - Mach

---

- Mach communication is message based
  - Even system calls are messages
  - Each task gets two mailboxes at creation- Kernel and Notify
  - Only three system calls needed for message transfer  
`msg_send()` , `msg_receive()` , `msg_rpc()`
  - Mailboxes needed for communication, created via  
`port_allocate()`
  - Send and receive are flexible, for example four options if mailbox full:
    - ▶ Wait indefinitely
    - ▶ Wait at most n milliseconds
    - ▶ Return immediately
    - ▶ Temporarily cache a message





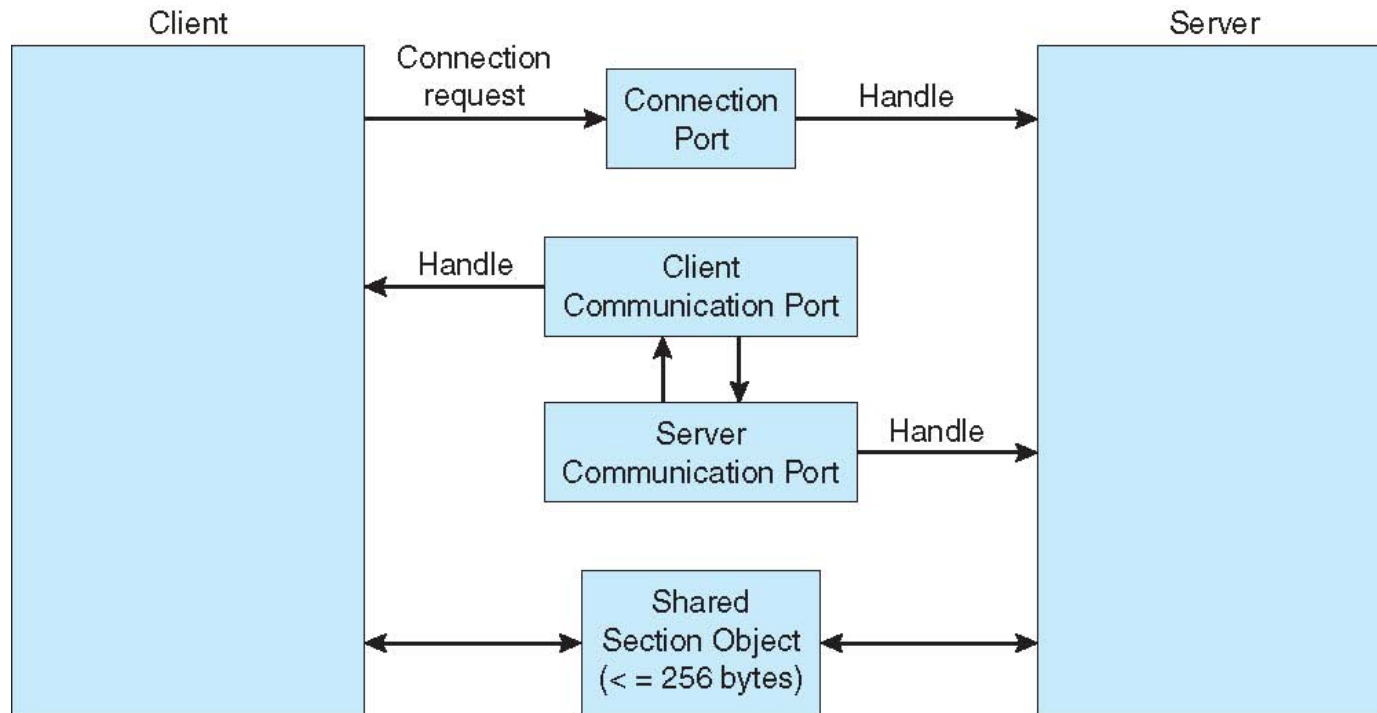
# Examples of IPC Systems – Windows

- Message-passing centric via **advanced local procedure call (LPC)** facility
  - Only works between processes on the same system
  - Uses ports (like mailboxes) to establish and maintain communication channels
  - Communication works as follows:
    - ▶ The client opens a handle to the subsystem's **connection port** object.
    - ▶ The client sends a connection request.
    - ▶ The server creates two private **communication ports** and returns the handle to one of them to the client.
    - ▶ The client and server use the corresponding port handle to send messages or callbacks and to listen for replies.





# Local Procedure Calls in Windows





# Communications in Client-Server Systems

---

- Sockets
- Remote Procedure Calls
- Pipes
- Remote Method Invocation (Java)





# Sockets

---

- A **socket** is defined as an endpoint for communication
- Identified by concatenation of IP address and **port** – a number included at start of message packet to differentiate network services on a host
- The socket **161.25.19.8:1625** refers to port **1625** on host **161.25.19.8**
- Communication consists between a pair of sockets
- **Note!** All ports below 1024 are well known, used for standard services (Eg., Telnet server listens to port 23, an FTP server listens to port 21, a web or HTTP server listens to port 80)





# Sockets

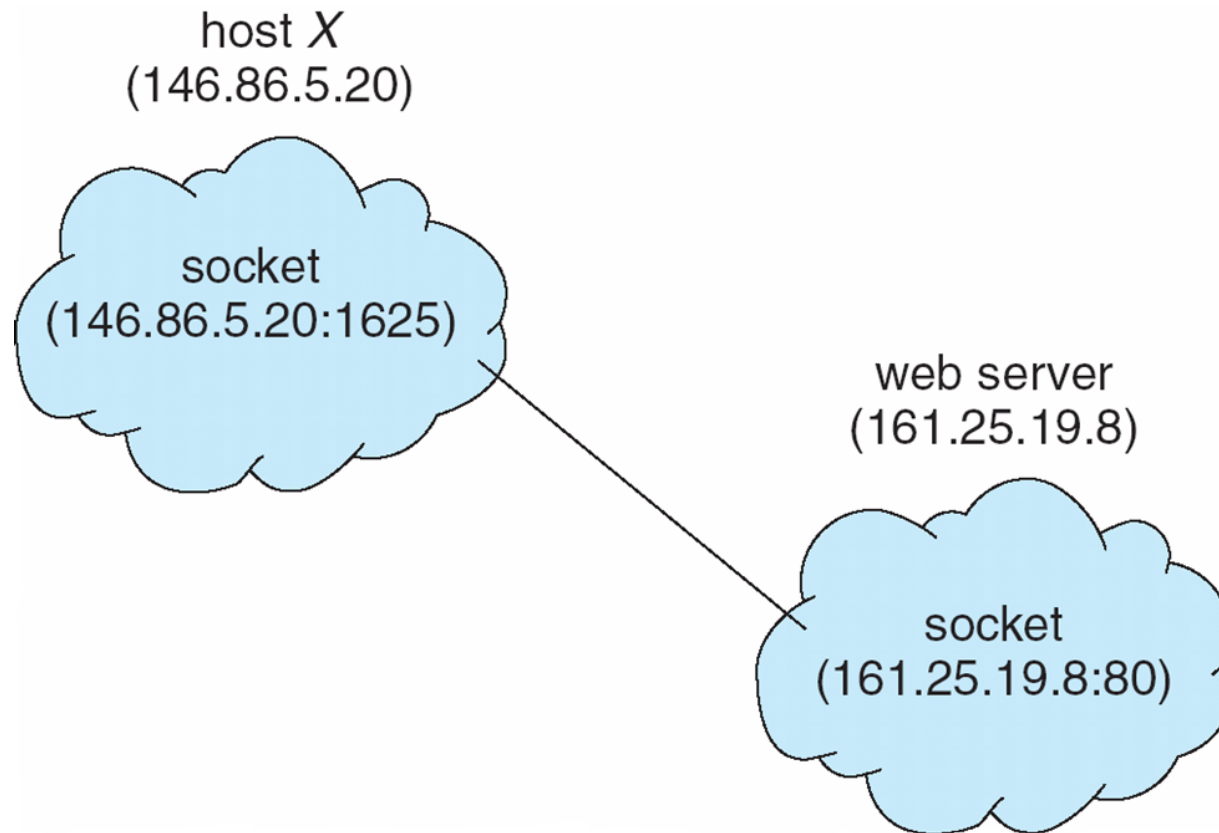
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- Sockets Example: How the Web Works
  - A web, or HTTP server, listens to port 80.
  - A client [process] initiates a request for connection, host computer assigns an arbitrary port number greater than 1024 .
  - The connection will consist of a pair of sockets
  - The packets traveling between hosts are delivered to the appropriate process based on the destination port number





# Socket Communication





# Sockets

---

- All connections must be unique (ie, all connections consist of a unique pair of sockets)
- The IP address 127.0.0.1 is a special IP address known as the loopback (ie, this refers to “self”)
- This allows a client and a server on the same host to communicate using TCP/IP protocol





# Sockets

---

- Steps in Establishing a Socket on the Client:
  1. Create a socket with the `socket()` system call
  2. Connect the socket to the address of the server using the `connect()` system call
  3. Send and receive data using `send()` and system calls.
  4. Release kernel data structures using `close()`





# Sockets

---

## ■ Steps in Establishing a Socket on the Server:

1. Create a socket with the `socket()` system call
2. Bind the socket to an address using the `bind()` system call. For a server socket on the Internet, an address consists of a port number on the host machine.
3. Listen for connections with the `listen()` system call.
4. Establish a connection with a specific client using the `accept()` system call. This call typically blocks until a client connects with the server.
5. Send and receive data using `send()` and system calls
6. Release kernel data structures using `close()`





# Sockets in Java

- Three types of sockets
  - **Connection-oriented (TCP)**
  - **Connectionless (UDP)**
  - **MulticastSocket** class— data can be sent to multiple recipients
- Consider this “Date” server:

```
import java.net.*;
import java.io.*;

public class DateServer
{
 public static void main(String[] args) {
 try {
 ServerSocket sock = new ServerSocket(6013);

 /* now listen for connections */
 while (true) {
 Socket client = sock.accept();

 PrintWriter pout = new
 PrintWriter(client.getOutputStream(), true);

 /* write the Date to the socket */
 pout.println(new java.util.Date().toString());

 /* close the socket and resume */
 /* listening for connections */
 client.close();
 }
 }
 catch (IOException ioe) {
 System.err.println(ioe);
 }
 }
}
```





# Remote Procedure Calls

---

- Remote procedure call (RPC) abstracts procedure calls between processes on networked systems
  - Again uses ports for service differentiation
- **Stubs** – client-side proxy for the actual procedure on the server
- The client-side stub locates the server and **marshalls** the parameters
- The server-side stub receives this message, unpacks the marshalled parameters, and performs the procedure on the server
- On Windows, stub code compile from specification written in **Microsoft Interface Definition Language (MIDL)**





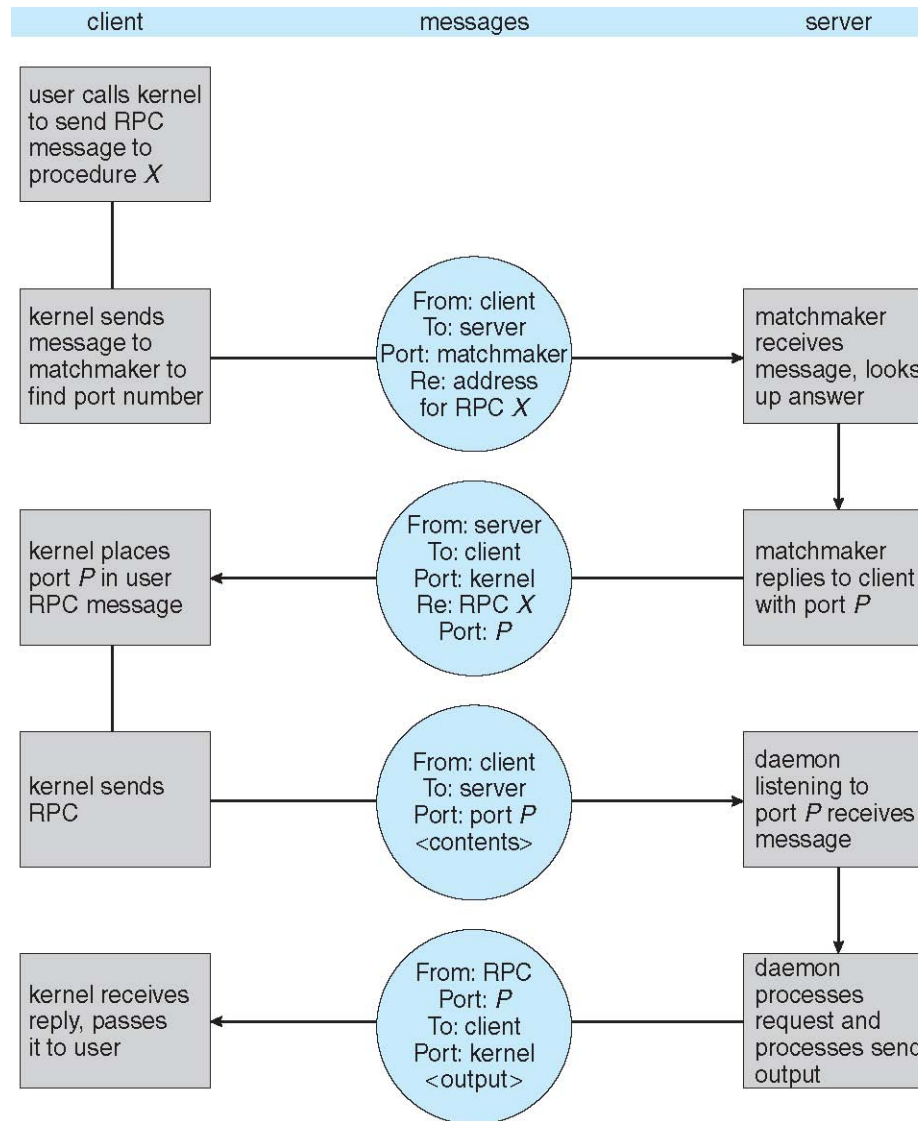
# Remote Procedure Calls (Cont.)

- Data representation handled via **External Data Representation (XDL)** format to account for different architectures
  - **Big-endian** and **little-endian**
- Remote communication has more failure scenarios than local
  - Messages can be delivered ***exactly once*** rather than ***at most once***
- OS typically provides a rendezvous (or **matchmaker**) service to connect client and server





# Execution of RPC





# Pipes

---

- Acts as a conduit allowing two processes to communicate
- Issues:
  - Is communication unidirectional or bidirectional?
  - In the case of two-way communication, is it half or full-duplex?
  - Must there exist a relationship (i.e., ***parent-child***) between the communicating processes?
  - Can the pipes be used over a network?





# Ordinary Pipes

- Ordinary Pipes allow communication in standard producer-consumer style
  - Producer writes to one end (the **write-end** of the pipe)
  - Consumer reads from the other end (the **read-end** of the pipe)
    - ▶ Ordinary pipes are therefore unidirectional
- Require parent-child relationship between communicating processes
- Ordinary pipes – cannot be accessed from outside the process that created it. Typically, a parent process creates a pipe and uses it to communicate with a child process that it created.





# Named Pipes

---

- Named Pipes are more powerful than ordinary pipes
- Communication is bidirectional
- No parent-child relationship is necessary between the communicating processes
- Once established, several processes can use the named pipe for communication
- Continues to exist after communicating processes have finished



# End of Chapter 3

---

